

Customer Churn Prediction and Risk Segmentation Using Machine Learning

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Abstract

This project focuses on predicting customer churn using the IBM Telco Customer Churn dataset and translating the results into actionable business insights. The main objective was to identify customers who are likely to leave so that the company can take preventive action before churn occurs. The dataset was cleaned and prepared through preprocessing steps such as handling missing or inconsistent values, simplifying selected categorical variables, encoding categorical features, and scaling numerical features where appropriate. Multiple classification machine learning models were developed and compared, including logistic regression, random forest, and gradient boosting (XGBoost). Model performance was evaluated using metrics such as ROC-AUC, accuracy, precision, recall, and F1-score, along with cross-validation on the training set. After selecting the best-performing model, feature importance was analyzed to identify the variables most strongly associated with churn. Predicted churn probabilities were also converted into low-, medium-, and high-risk groups to support customer prioritization. Finally, the results were presented in an executive-style Tableau dashboard designed to highlight churn drivers, risk segments, and business-focused insights for decision-making.